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$$\int \tan^2(4x)dx$$

$$t = 4x \Rightarrow dt = 4dx$$

$$= \frac{1}{4} \int \tan^2(t)dt = \frac{1}{4} \left(\int \sec^2(t)dt - \int dt \right)$$

$$= \frac{1}{4} \tan t - \frac{1}{4}t + C$$

$$= \frac{\tan(4x)}{4} - \frac{4}{4}x + C = \frac{\tan(4x)}{4} - x + C$$

References